AutoDSC: An Automated Data Stream Clustering Framework

Anonymous

***Abstract*—Data stream clustering faces unique challenges in dynamic environments where data distributions evolve continuously and computational resources are limited. While AutoML has transformed supervised learning, its application to unsupervised streaming tasks remains largely unexplored. Traditional clustering algorithms are sensitive to algorithm selection, hyperparameter configuration, and changes in data characteristics. We introduce AutoDSC (Automated Data Stream Clustering), a novel framework that automatically optimizes algorithm selection and hyperparameter tuning for real-time stream clustering. AutoDSC employs Population-Guided Stochastic Search Optimization (PGSSO), leveraging a recency-weighted elite archive to guide efficient parameter exploration while dynamically adjusting the exploration rate based on performance feedback. The framework adapts to concept drift implicitly: every window it re-evaluates the population and re-selects the incumbent, and its exploration rate rises automatically when clustering quality degrades. Robust performance assessment is provided by a multi-objective score combining three clustering validation indices. Our theoretical analysis establishes regret bounds for the optimization process. Comprehensive experiments demonstrate that AutoDSC achieves statistically significant improvements in clustering quality while maintaining computational efficiency compared to existing automated approaches.**

***Index Terms*—data stream clustering, AutoML, concept drift, hyperparameter optimization, online learning**

I. Introduction

Automated Machine Learning (AutoML) has significantly advanced the accuracy and efficiency of machine learning workflows by automating tasks such as algorithm selection, hyperparameter optimization, and pipeline synthesis [1]. While AutoML has seen widespread adoption in supervised learning, its application to clustering remains comparatively less explored, mainly due to the absence of a universally applicable objective function for evaluation [2]. Recent AutoML solutions for clustering include AutoML4Clust [3], AutoCluster [4], cSmartML [5], ML2DAC [6], TPE-AutoClust [7], and PoAC [8]. These approaches address clustering challenges using various automated methodologies such as meta-learning, pipeline composition, and problem-specific heuristics.

However, with the increasing availability of streaming data from sources like Internet of Things (IoT) sensors, web traffic, and social media, traditional static AutoML approaches are insufficient [9]. They cannot adapt to dynamic environments or process data in a single pass due to concept drift and memory limitations [10]. Existing online AutoML frameworks such as Online AutoML [11], ASML [12], ChaCha [13], and Auto-Reg [14] primarily focus on supervised learning tasks. Meanwhile, unsupervised learning, particularly clustering, requires specialized AutoML solutions. At present, ConfStream [15] and OnlineAutoClust [16] represent the only established AutoML solutions for stream clustering. Nevertheless, they suffer from significant limitations. ConfStream relies on single clustering evaluation metrics, and OnlineAutoClust uses computationally intensive offline search methods, which are inefficient for real-time streaming scenarios. These limitations highlight the critical need for tailored AutoML solutions designed specifically for data stream clustering [17]. This paper introduces AutoDSC (Automated Data Stream Clustering), a comprehensive AutoML framework for unsupervised data stream clustering. Our key contributions are:

- We propose AutoDSC, the first AutoML framework to unify recency-weighted elite-guided parameter optimization, implicit drift adaptation, and multi-objective clustering evaluation in a single practical framework, introducing Population-Guided Stochastic Search Optimization (PGSSO) as a novel optimization mechanism that dynamically balances exploration and exploitation based on performance feedback in non-stationary streaming environments.
- We establish formal regret bounds proving convergence for the optimization process and demonstrate efficient recovery after concept drift.
- Through extensive experiments on synthetic and real-world data streams, we demonstrate that AutoDSC achieves statistically significant improvements in clustering quality while maintaining computational efficiency.

II. Related Work

Data stream clustering addresses the challenges of processing continuous, high-velocity data with limited computational resources [10], [18]. CluStream [19] pioneered the micro-cluster approach using pyramidal time frames to maintain summary statistics for scalable analysis. DenStream [20] advanced density-based clustering by introducing core-micro-clusters, potential micro-clusters, and outlier micro-clusters. DBSTREAM [21] enhanced density estimation by incorporating shared density graphs that explicitly model inter-micro-cluster relationships. This provides better resilience to concept drift compared to DenStream. For k-means-based approaches [22], mini-batch optimization combined with sparse projections has been developed to enable web-scale clustering with significantly reduced latency for real-time streaming applications. STREAMKMeans [23] is an alternative version of the stream local search [24] that replaces k-medians with incremental k-means.

Recent developments have focused on automated algorithm selection and hyperparameter optimization (CASH) [1] for stream clustering. ConfStream [15] represents an early attempt at automated stream clustering configuration, using an ensemble of different clustering algorithms and configurations that dynamically adapt by evaluating clustering quality and sampling nearby new configurations proportionally to their performance using a normal distribution. However, ConfStream has significant limitations, including its reliance on a single Silhouette Width metric for quality assessment and limited parameter search capabilities. OnlineAutoClust [16] developed an AutoML framework that combines Bayesian optimization with clustering ensembles, employing Tree-structured Parzen Estimators (TPE) for algorithm selection and hyperparameter tuning. However, OnlineAutoClust suffers from critical limitations, including dependence on offline tuning methods, high computational requirements, and sensitivity to allocated time budgets, which can lead to suboptimal pipeline selections during rapid concept drift. In contrast to these approaches, AutoDSC adapts to drift implicitly through continual per-window re-evaluation and an adaptive exploration rate, avoiding both single-metric evaluation and expensive offline optimization.

III. Automated Data Stream Clustering

We introduce Automated Data Stream Clustering (AutoDSC), a novel AutoML framework tailored for non-stationary data stream clustering. AutoDSC processes data in fixed-size windows. At the end of each window, AutoDSC evaluates clustering configurations using multiple normalized validation indices. The framework maintains a recency-weighted elite archive of high-performing configurations to guide future parameter exploration via PGSSO. It adaptively balances exploration and exploitation through dynamic rate adjustment based on performance feedback. Adaptation to distributional change is implicit: each window the population is re-evaluated and the incumbent re-selected, and a drop in clustering quality automatically raises the exploration rate so new configurations are sought for the new regime. Figure 1 illustrates the high-level workflow of AutoDSC, while Algorithm 1 details the complete operational steps.

AutoDSC processes an infinite data stream $\mathcal{S} = \{x_1, x_2, \ldots, x_t, \ldots\}$ where each instance $x_t \in \mathbb{R}^d$ arrives at timestamp t through a fixed-size windowing mechanism $W_t = \{x_{t-w+1}, x_{t-w+2}, \ldots, x_t\}$ with capacity w. The framework maintains a configuration population $\mathcal{A}_t$ containing multiple clustering algorithms with different hyperparameter settings, continuously adapts these configurations based on performance feedback, and provides real-time predictions using the best-performing configuration. The configuration space $\Theta = \prod_{i=1}^{|\mathcal{A}|} \Theta_i$ encompasses parameter spaces across all algorithms in the portfolio $\mathcal{A}$, where each configuration $\theta = (a_i, \phi_i)$ specifies both algorithm selection $a_i \in \mathcal{A}$ and hyperparameter values $\phi_i \in \Theta_i$. As each new data point arrives, the system performs incremental updates across all configurations:

$$\forall \theta_i \in \mathcal{A}_t : \quad M_i^{(t)} = \text{LearnOne}(M_i^{(t-1)}, x_t) \tag{1}$$

where $M_i^{(t)}$ represents the model state of configuration θ_i at time t. When the window reaches capacity, the framework initiates evaluation and optimization operations.

When the window fills, AutoDSC evaluates clustering quality using three normalized clustering validation indices: Normalized Silhouette Coefficient NSC $= (1 + \text{SC})/2$, Normalized Davies-Bouldin Index NDBI $= 1 - \tanh(\text{DBI})$, and Normalized Calinski-Harabasz Index NCHI $= \tanh(\log(1 + \text{CHI})/10)$, ensuring all metrics are bounded to $[0, 1]$ where higher values indicate better clustering quality. Each configuration is scored on the entire window (the per-point cluster assignments produced during LearnOne are retained for this purpose), so no sub-sampling is required. The expected clustering quality for configuration θ at time t is:

$$\mu_t(\theta) = \frac{1}{3}[\text{NSC} + \text{NDBI} + \text{NCHI}] \tag{2}$$

AutoDSC dynamically adjusts exploration behavior based on performance improvement trends. The exploration rate α_t adapts based on relative improvement:

$$\Delta\mu_t = \mu_t(\theta_t^*) - \mu_{t-1}(\theta_{t-1}^*) \tag{3}$$

$$\alpha_{t+1} = \begin{cases} \max(0.1,\ \alpha_t \cdot (1 - \beta_t)) & \text{if } \Delta\mu_t > 0 \\ \min(1.0,\ \alpha_t \cdot (1 + \beta_t)) & \text{if } \Delta\mu_t \leq 0 \end{cases} \quad \beta_t = \min(\gamma, |r_t|) \tag{4}$$

where $r_t = \Delta\mu_t / |\mu_{t-1}|$ is the relative improvement and β_t is the resulting step, capped at $\gamma = 0.5$ for stability. This parameter-free mechanism decreases exploration when performance improves (favoring exploitation) and increases it when performance stagnates or drops—as happens under concept drift—encouraging broader search to recover.

The framework employs **Population-Guided Stochastic Search Optimization (PGSSO)** (Algorithm 2) to jointly optimize algorithm selection and hyperparameter tuning. PGSSO maintains an elite archive $\mathcal{E}_t = \{(\theta_i, s_i)\}_{i=1}^{e}$ where each configuration $\theta_i = (a_i, \phi_i)$ pairs an algorithm choice with its hyperparameters and $s_i = \mu_t(\theta_i)$ is its score. To remain responsive to drift, elites are ranked by a *recency-weighted* score $s_i/(1 + g - g_i)$, where g is the current generation and g_i the generation at which θ_i was last scored; stale high-scoring configurations are thus down-weighted in favor of recently effective ones. By tracking which algorithm-hyperparameter combinations perform best, PGSSO naturally learns both which algorithms suit current data characteristics and their optimal settings. The elite population encodes this joint preference: frequently appearing algorithms indicate superior algorithmic choices, while their parameter distributions reveal effective settings. PGSSO adaptively balances exploration and exploitation through dynamic rate α_t, computing statistical distributions over both algorithm selections (via

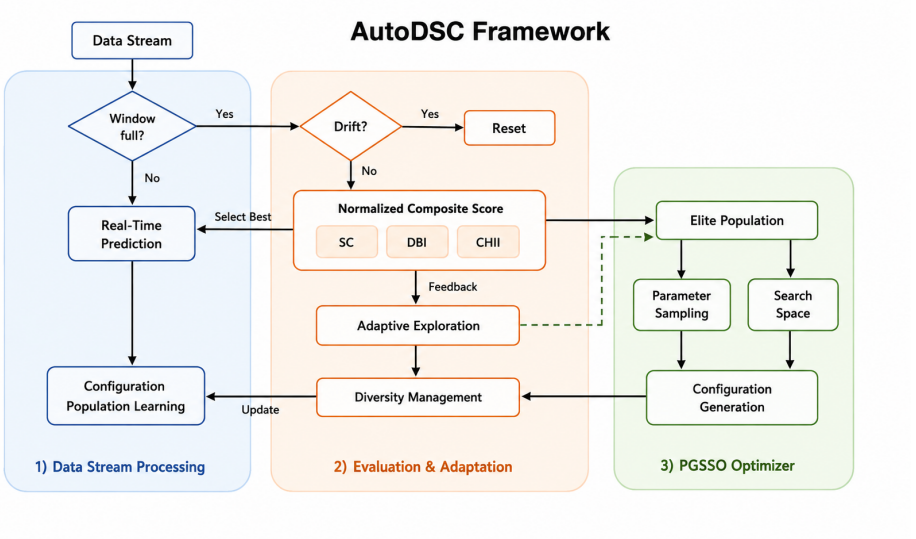


Fig. 1. AutoDSC Framework Overview

frequencies) and their parameters (via means μ_j and variances σ_j^2), which together guide sampling of new promising algorithm-hyperparameter configurations.

PGSSO generates new configurations through probabilistic sampling:

$$\theta_{\text{new}} = \begin{cases} \text{Elite-Guided}(\mathcal{E}_t) & \text{with probability } (1-\alpha_t) \\ \text{Random}(\Theta) & \text{with probability } \alpha_t \end{cases} \tag{5}$$

Elite-guided sampling first selects an algorithm based on elite frequencies, then samples its hyperparameters from learned distributions. Algorithm selection uses:

$$P(a = a_k) = \frac{\sum_{(\theta,s)\in\mathcal{E}_t} \mathbf{I}[a = a_k]}{|\mathcal{E}_t|} \tag{6}$$

where $a_k \in \mathcal{A}$ represents available algorithms. Given selected algorithm a_k, numerical hyperparameters use Gaussian distributions from elite configurations of that algorithm:

$$\phi_j \sim \mathcal{N}(\mu_j, \sigma_j^2) \text{ where } \mu_j = \frac{1}{|\mathcal{E}_t^{a_k}|} \sum_{(\theta,s)\in\mathcal{E}_t^{a_k}} \phi_j \tag{7}$$

and $\mathcal{E}_t^{a_k} = \{(\theta, s) \in \mathcal{E}_t : a = a_k\}$ contains elite configurations using algorithm a_k. Categorical hyperparameters similarly use frequency-based sampling within algorithm-specific elites. Figure 2 illustrates this hierarchical sampling process.

Population diversity is maintained by computing configuration distances:

$$d(\theta_1, \theta_2) = \begin{cases} 1.0 & \text{if algorithms differ} \\ \frac{1}{|\mathcal{P}|} \sum_{p\in\mathcal{P}} d_p(\theta_1[p], \theta_2[p]) & \text{otherwise} \end{cases} \tag{8}$$

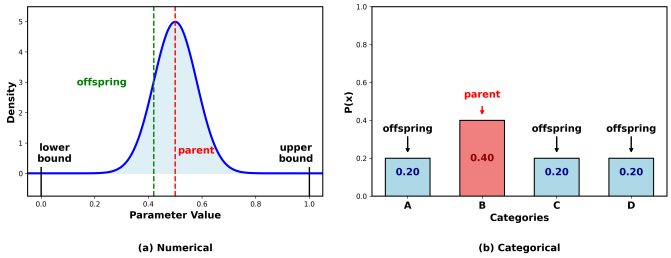


Fig. 2. Parameter Sampling Visualization.

where $\mathcal{P}$ represents common parameters and d_p computes normalized parameter-specific distances. When the population exceeds size K, it is pruned by first retaining the incumbent and top-scoring configurations and then greedily adding the remaining configuration that is *farthest* (by d) from those already kept—a farthest-first selection that preserves diversity without requiring a fixed acceptance threshold.

Throughout the process, AutoDSC provides real-time clustering predictions using the incumbent (best) configuration:

$$\theta_t^* = \arg\max_{\theta_i\in\mathcal{A}_t} \mu_t(\theta_i) \tag{9}$$

A. Computational Complexity

The framework's computational complexity scales efficiently with population size and window parameters. Per-instance processing requires $O(K \cdot C)$ operations where $K = |\mathcal{A}_t|$ and C denotes average algorithm complexity. Window-end evaluation involves $O(K \cdot w \cdot d)$ operations to score each configuration on the full window. Total per-window complexity is $O(K \cdot w \cdot d + K^2)$, where K^2 accounts for diversity-based population management. Crucially, K is fixed, so per-point cost does not grow with stream length. Space complexity is $O((K + e_{\max}) \cdot |\Theta| + w \cdot d)$.

Algorithm 1 AutoDSC

Input: Data stream $\mathcal{S} = \{x_1, x_2, \ldots\}$, $x_t \in \mathbb{R}^d$
Parameter: Window size w, Population size K
Output: Cluster predictions $\{y_t\}_{t=1}^{\infty}$ using θ_t^*

$e_{\max} \leftarrow \lfloor K/2 \rfloor$
$\mathcal{E}_0 \leftarrow \emptyset$, $\alpha_0 \leftarrow 0.5$
$\mathcal{A}_0 \leftarrow \{\theta_i\}_{i=1}^{K}$ where $\theta_i \sim \text{Uniform}(\Theta)$
$W_0 \leftarrow \emptyset$, $\theta_0^* \leftarrow \emptyset$
$\mu_{-1}(\theta_{-1}^*) \leftarrow -\infty$
for each $x_t \in \mathcal{S}$ **do**
  $W_t \leftarrow W_{t-1} \cup \{x_t\}$
  for $\theta_i \in \mathcal{A}_t$ **do**
    $\ell_{i,t} \leftarrow \textsc{PredictOne}(M_i, x_t)$ ▷ store assignment
    $M_i^{(t)} \leftarrow \textsc{LearnOne}(M_i^{(t-1)}, x_t)$
  end for
  if $|W_t| = w$ **then**
    for $\theta_i \in \mathcal{A}_t$ **do**
      $L_i \leftarrow \{\ell_{i,\tau} : x_\tau \in W_t\}$
      $\mu_t(\theta_i) \leftarrow \textsc{NormCVI}(W_t, L_i)$
    end for
    $\mu_{\text{best}}^{(t)} \leftarrow \max_i \mu_t(\theta_i)$
    $\alpha_{t+1} \leftarrow \textsc{Adapt}(\alpha_t, \mu_{t-1}(\theta_{t-1}^*), \mu_{\text{best}}^{(t)})$
    $\mathcal{C}_{\text{new}} \leftarrow \text{PGSSO}(\{(\theta_i, \mu_t(\theta_i))\}, \alpha_{t+1}, \Theta)$
    $\mathcal{A}_{t+1} \leftarrow \textsc{SelectDiverse}(\mathcal{A}_t \cup \mathcal{C}_{\text{new}}, K)$
    $\theta_t^* \leftarrow \arg\max_{\theta_i \in \mathcal{A}_{t+1}} \mu_t(\theta_i)$
    $W_t \leftarrow \emptyset$
  end if
end for

Algorithm 2 PGSSO

Input: Current configurations with scores $\{(\theta_i, s_i)\}$
Parameter: Exploration rate $\alpha \in [0, 1]$, Parameter space Θ
Output: New configuration set $\mathcal{C}_{\text{new}}$

$\mathcal{E} \leftarrow \textsc{UpdateElitePopulation}(\{(\theta_i, s_i)\}, e_{\max})$
$\mathcal{E} \leftarrow \textsc{UpdateParameterDistributions}(\mathcal{E})$
$\mathcal{C}_{\text{new}} \leftarrow \emptyset$
for $j = 1$ to n_{gen} **do**
  $u \leftarrow \text{Uniform}(0, 1)$
  if $u > \alpha$ **then**
    $\theta_{\text{new}} \leftarrow \textsc{SampleFromElite}(\mathcal{E}, \Theta)$
  else
    $\theta_{\text{new}} \leftarrow \textsc{RandomSample}(\Theta)$
  end if
  $\mathcal{C}_{\text{new}} \leftarrow \mathcal{C}_{\text{new}} \cup \{\theta_{\text{new}}\}$
end for
return $\mathcal{C}_{\text{new}}$

IV. Theoretical Analysis

We analyze AutoDSC using a phase-based approach where phases correspond to periods with fixed incumbent configurations. Let $j \in \{1, ..., T/w\}$ index windows where T/w represents the total number of windows given T instances and window size w, and $t \in \{1, ..., T\}$ index instances. We analyze AutoDSC's performance over T/w windows divided into $M + 1$ phases, where phase m spans N_m windows from window j_m to $j_{m+1} - 1$ with population $\mathcal{A}_{j_m}$ and incumbent configuration $\theta_{j_m}^*$. We track cumulative evaluations $n_{\theta,j}$ for each configuration θ across all windows up to window j. The expected clustering quality for configuration θ at window j is defined as:

$$\mu_j(\theta) = \mathbb{E}[r_j(\theta)] \tag{10}$$

where $r_j(\theta) \in [0, 1]$ represents the instantaneous composite CVI score, and $\theta_j^{\text{opt}} = \arg\max_{\theta \in \Theta} \mu_j(\theta)$ denotes the optimal configuration at window j.

Definition 1: Let $p \in (0, 1)$ denote the sample complexity parameter where estimation error decreases as $O(n^{-p})$ with n samples (typically $p = 1/2$ for clustering).

Definition 2: Let comp_θ denote the computational complexity of evaluating configuration θ on one window.

Assumption 1: Rewards are bounded: $r_j(\theta) \in [0, 1]$.

Assumption 2: Within stationary segments, $\mu_j(\theta)$ is L-Lipschitz in θ.

Assumption 3: The optimal configuration θ_j^{opt} enters the population within $O(\log |\Theta|)$ windows after each concept drift.

Define confidence intervals:

$$\epsilon_{\theta,j} = O\left(\sqrt{\frac{\log(|\Theta|j/\delta)}{n_{\theta,j}}}\right) \tag{11}$$

where $n_{\theta,j}$ is the number of evaluations of θ by window j and $\delta \in (0, 1)$ is a confidence parameter [25].

Lemma 1 (PGSSO Selection Properties): With probability $\geq 1 - \delta$ [26]:

Claim 1: If $\mu_j(\theta) - \mu_j(\theta_{j_m}^*) + 2\epsilon_{\theta,j} + 3\epsilon_{\theta_{j_m}^*,j} > 0$, then PGSSO selects θ as the new incumbent $\theta_{j_{m+1}}^*$.

Claim 2: When the incumbent switches from $\theta_{j_m}^*$ to $\theta_{j_{m+1}}^*$, we have $\mu_{j_{m+1}}(\theta_{j_m}^*) - \mu_{j_{m+1}}(\theta_{j_{m+1}}^*) > \epsilon_{\theta_{j_m}^*,j_{m+1}}$.

Claim 3: If $\mu_j(\theta) < \mu_j(\theta_{j_m}^*)$, then θ will not replace $\theta_{j_m}^*$ as the incumbent.

Lemma 2 (Drift Recovery via Adaptive Exploration): When a concept drift degrades incumbent quality ($\Delta\mu_j \leq 0$), the exploration rate increases (Eq. 4). Under Assumption 3, the post-drift optimal configuration re-enters the population within $O(\log |\Theta|)$ windows with probability $\geq 1 - \delta$, without any explicit drift detector.

Lemma 3 (Adaptive Exploration): The adaptive exploration mechanism in PGSSO, which adjusts α_j based on performance trends, improves the expected regret constants by a factor of $(1+\gamma)^{-1}$ compared to fixed exploration, where γ is the bounded adaptation gain (the per-step exploration adjustment is capped at $\gamma = 0.5$).

Proposition 1 (Window Regret): With sample complexity parameter p and properly set confidence intervals, AutoDSC achieves:

$$\sum_{m=0}^{M} \sum_{j=j_m}^{j_{m+1}-1} \left(\mu_j(\theta_j^{\text{opt}}) - \mu_j(\theta_{j_m}^*)\right)$$
$$= \tilde{O}\left(\frac{K \cdot \text{comp}_{\theta^{\text{opt}}} \cdot (T/w)^p}{p(1+\gamma)} + \text{comp}_{\theta^*}(T/w)^{\frac{1}{2-p}}\right) \quad (12)$$

where $K = \max_j |\mathcal{A}_j|$ is the maximum population size.

Proof Sketch: The bound decomposes regret into two terms: (i) regret from suboptimal population sampling, bounded by PGSSO's exploration guarantee with adaptive exploration reducing the constant by $(1+\gamma)^{-1}$, and (ii) regret from incumbent updates. Lemma 1 ensures incumbent switches occur only when sufficiently beneficial, limiting the number of phases $M = O((T/w)^{\frac{1-p}{2-p}})$. The confidence intervals shrink as $O(n_{\theta,j}^{-1/2})$, yielding the stated bounds [25].

Theorem 1 (Total Regret): Under the same conditions, AutoDSC achieves instance-level regret:

$$\sum_{t=1}^{T} \left(\mu_{\lceil t/w \rceil}(\theta^{\text{opt}}_{\lceil t/w \rceil}) - \mu_{\lceil t/w \rceil}(\theta^*_{\lceil t/w \rceil})\right)$$
$$= \tilde{O}\left(\frac{K \cdot \text{comp}_{\theta^{\text{opt}}} \cdot T^p}{(1+\gamma)} + \text{comp}_{\theta^*} T^{\frac{1}{2-p}}\right) \quad (13)$$

Proof Sketch: Since clustering quality is constant within windows, summing over T instances with window-level quality yields:

$$\sum_{t=1}^{T}(\cdot) = \sum_{j=1}^{T/w} w \cdot (\mu_j(\theta_j^{\text{opt}}) - \mu_j(\theta_j^*)) \quad (14)$$

Applying Proposition 1 and noting w is constant gives the result.

Corollary 1 (Drift Recovery): After concept drift, AutoDSC recovers ϵ-optimal performance within $O(\log|\Theta|/\epsilon^2)$ windows.

For $p = 1/2$, the regret bound becomes $\tilde{O}(\sqrt{T})$, matching optimal rates for non-stationary bandits with concept drift [27].

V. Experimental Setup

Our experimental design follows rigorous evaluation protocols to ensure reproducible and statistically valid results.[1]

Datasets: We selected 10 datasets comprising 5 real-world and 5 synthetic streams with varying sizes, dimensionalities, and cluster counts, as summarized in Table I.

Evaluation Protocol: All experiments use prequential evaluation (test-then-train) [10] with 10 independent runs using random seeds 1-10 to account for stochastic variability. Clustering quality is measured using the Adjusted Rand Index (ARI) [28] and CMM [29], where ground truth labels are used solely for evaluation, not training or model selection. We also record throughput (instances/s), per-point latency (μs), and peak memory usage (MB) to assess computational efficiency. Statistical significance is assessed using the Friedman test followed by the Nemenyi post-hoc test [30] with $\alpha = 0.05$.

[1]Datasets and code: https://anonymous.4open.science/r/AutoDSC

TABLE I
Summary of Datasets

Dataset Name	Type	Instances	Attributes	Clusters
Insects	Real	52,848	33	6
KDD99	Real	100,655	38	12
PenDigits	Real	10,992	16	10
Shuttle	Real	58,000	9	7
Vehicle	Real	98,528	100	3
Expanding	Synth	100,000	5	5
Gradual	Synth	100,000	5	5
Merge-Split	Synth	100,000	5	5
Sudden	Synth	100,000	5	5
Fast-RBF	Synth	100,000	5	5

Baseline Methods: We compare AutoDSC against eight baselines spanning three categories. (i) *AutoML stream-clustering methods*: *ConfStream* [15] and *OnlineAutoClust* [16], established frameworks that optimize hyperparameters online, and *RandomSearch* [31], which shares AutoDSC's framework structure but generates configurations by purely random sampling—providing a direct comparison to our guided optimization. (ii) The *rule-based adaptive* method *MOStream* [32]. (iii) *Single fixed algorithms*—KMeans, CluStream, DBSTREAM, and DenStream—run with their library-default parameters, reflecting common practice among non-expert users who apply standard algorithms without tuning.

Configuration: Unless otherwise stated, AutoDSC uses its default configuration: population size $K = 10$ and window size $w = 1000$, with the exploration rate adapting automatically from an initial value of 0.5 (Section VI-A analyzes sensitivity to K and w). The AutoML baselines search over the same algorithm portfolio and hyperparameter ranges specified in Table II; the single fixed algorithms use their library defaults. Other baseline-specific parameters remain as reported in their original papers.

All experiments were conducted on AMD EPYC 7702 64-core servers with 1TB RAM, utilizing parallelization for computational efficiency. All clustering algorithms were implemented using the River Python library [23] for online machine learning.

VI. Results Analysis

Table III presents a comprehensive comparison of AutoDSC against eight baselines across ten diverse real-world and synthetic datasets, evaluating clustering quality with both ARI and CMM. All baselines use their library-default hyperparameters; AutoDSC selects them automatically.

AutoDSC achieves the highest overall mean ARI of **0.622**, ahead of every baseline—surpassing the strongest AutoML competitor ConfStream (0.572) by 5.0 points, OnlineAutoClust (0.558) by 6.4 points, and RandomSearch (0.534) by 8.8 points—and it attains the best mean rank on both ARI and CMM. AutoDSC ranks first on ARI for 7 of the 10 datasets and is never worse than second, with the largest margins on the drifting synthetic streams (e.g. Fast-RBF, Gradual, Expanding all ≥ 0.99) and on the harder real stream PenDigits

TABLE II
SEARCH SPACE CONFIGURATION. WINDOW-DEPENDENT RANGES (TIME_WINDOW, CHUNK_SIZE) ARE SHOWN FOR THE DEFAULT WINDOW SIZE $w = 1000$.

Algorithm	Hyperparameter	Type	Range/Options
CluStream	n_macro_clusters	integer	[2, 15]
	max_micro_clusters	integer	[20, 100]
	time_window	integer	[250, 1500]
	micro_cluster_r_factor	numeric	[1.0, 3.0]
DenStream	decaying_factor	numeric	[0.01, 0.2]
	beta	numeric	[0.2, 0.8]
	mu	numeric	[2.0, 8.0]
	epsilon	numeric	[0.01, 1.0]
DBSTREAM	clustering_threshold	numeric	[0.1, 2.0]
	fading_factor	numeric	[0.001, 0.05]
	intersection_factor	numeric	[0.1, 0.8]
	minimum_weight	numeric	[0.5, 5.0]
STREAMKMeans	chunk_size	integer	[100, 500]
	n_clusters	integer	[2, 20]
	halflife	numeric	[0.05, 0.8]

(0.510 vs. 0.412 for the best baseline). Figure 3 shows the prequential trajectories: AutoDSC stays at or above all baselines throughout the stream on both a real (Pendigits) and a synthetic (Expanding) workload, reflecting rapid adaptation as the stream evolves.

To assess these performance gains statistically, the Critical Difference (CD) diagrams in Figure 4 rank all methods using the Friedman test with the Nemenyi post-hoc procedure ($\alpha = 0.05$). AutoDSC attains the best (lowest) mean rank on both ARI (1.4) and CMM, ahead of every baseline. We note that with ten datasets the Nemenyi critical difference is wide, so AutoDSC's rank advantage over the strongest adaptive baselines (ConfStream, OnlineAutoClust) is not separated by a statistically significant margin; nonetheless, AutoDSC ranks first on both metrics and is significantly better than the single fixed algorithms.

Beyond clustering quality, AutoDSC remains practical in throughput and memory. Among the AutoML methods it is the fastest, processing **2.8× more instances/s** than ConfStream and **2.0× more** than OnlineAutoClust, while using less memory than both (≈25 MB vs. 27 and 53 MB). The single fixed algorithms are naturally faster as they perform no search, and RandomSearch trades quality for speed; AutoDSC instead occupies the favorable region of the throughput–memory plane (Figure 5), pairing the highest clustering quality with a low memory footprint. OnlineAutoClust shows the worst footprint (53 MB) and MOStream the highest (166 MB).

A common concern for AutoML over streams is that search overhead accumulates as the stream lengthens. Figure 6 shows this does not occur: because AutoDSC maintains a fixed-size population, its per-point latency stays bounded across the entire stream—flat or decreasing as the population stabilizes—confirming real-time viability.

To quantify the contribution of each component, we ran a leave-one-out ablation over all real-world datasets (×10 seeds), disabling one mechanism at a time: recency-weighted elite ranking, the adaptive exploration rate, the multi-objective (3-CVI) score, and diversity-preserving pruning. (Replacing the guided PGSSO search by random configuration generation is already covered by the RandomSearch baseline.) Figure 7 reports each variant's score relative to the full model. The multi-objective CVI score is the most impactful component on real-world data—removing it lowers ARI by ∼2% and CMM by ∼3%—confirming that combining three complementary indices is more robust than any single index. Recency weighting and diversity pruning each give smaller but consistent gains, while the adaptive exploration rate has a negligible standalone effect on these streams. The full configuration is best overall, validating the integrated design.

Finally, AutoDSC adaptively reshapes its population of candidate algorithms as the stream evolves. Figure 8 shows the algorithm composition of the population across stream windows on the Insects dataset: each cell is the base algorithm occupying a population slot at that window. AutoDSC continuously maintains a diverse mix and shifts the composition between CluStream, DenStream, DBSTREAM and StreamKMeans as the active concept changes, rather than committing to a single algorithm.

A. Sensitivity Analysis

We analyze AutoDSC's sensitivity to its two free hyperparameters—window size w and population size K—averaged over all datasets (×10 seeds), jointly reporting clustering quality (ARI) and throughput (Figure 9). Window size shows a clear quality optimum at $w = 1000$: smaller windows give insufficient data for reliable per-window CVI evaluation, while larger windows delay adaptation to drift and reduce throughput. Population size exhibits a quality plateau—ARI rises steeply from $K = 5$ and saturates by $K \approx 15$—while throughput falls monotonically as K grows. The knee of this trade-off is at $K = 10$, which attains near-peak quality at substantially higher throughput, justifying it as the default. Overall, AutoDSC is stable within reasonable ranges and degrades gracefully outside them, with the default configuration ($w = 1000$, $K = 10$) sitting at the quality/efficiency sweet spot of both parameters.

VII. DISCUSSION

AutoDSC represents a significant advancement in AutoML for streaming environments through several key innovations. Our multi-objective evaluation strategy, combining three normalized clustering validation indices, provides robust performance assessment compared to single-metric approaches used in ConfStream (Silhouette Width only) and OnlineAutoClust's ensemble voting. This prevents optimization convergence to configurations that excel in one aspect while failing catastrophically in others [33], [34]. The PGSSO outperforms existing methods by dynamically adjusting exploration rates based on performance trends, enabling rapid parameter space exploration after concept drift while efficiently exploiting successful configurations during stable periods, contrasting with ConfStream's fixed ensemble strategy and OnlineAutoClust's

TABLE III

CLUSTERING QUALITY ACROSS DATASETS, AVERAGED OVER 10 SEEDS. FOR EACH METHOD, ARI AND CMM ARE REPORTED (↑ HIGHER IS BETTER). BEST PER METRIC PER ROW IN **BOLD**, SECOND BEST UNDERLINED. METHODS GROUPED INTO AUTOML, THE RULE-BASED ADAPTIVE METHOD MOSTREAM, AND SINGLE (FIXED) ALGORITHMS. CONFSTR=CONFSTREAM, RANDS=RANDOMSEARCHSTREAM, OACLE=ONLINEAUTOCLUSTENSEMBLE, CLUSTR=CLUSTREAM, DENSTR=DENSTREAM.

Dataset	AutoML								Rule-based		Single algorithm							
	AutoDSC		ConfStr		RandS		OACIE		MOStr		KMeans		CluStr		DBSTR		DenStr	
	ARI	CMM	ARI	CMM	ARI	CMM	ARI	CMM	ARI	CMM	ARI	CMM	ARI	CMM	ARI	CMM	ARI	CMM
Insects	**0.117**	0.905	0.016	0.827	0.054	0.880	0.048	**0.969**	0.090	0.964	0.023	0.907	0.100	0.947	-0.000	0.188	0.000	0.000
KDD99	**0.037**	0.942	**0.037**	0.910	0.033	0.935	0.036	0.626	0.018	0.780	0.032	0.641	0.035	0.890	0.029	0.986	**0.037**	**0.990**
PenDigits	**0.510**	**0.888**	0.271	0.668	0.367	0.814	0.311	0.777	0.363	0.764	0.032	0.514	0.319	0.738	0.412	0.823	0.000	0.000
Shuttle	0.363	0.900	0.341	0.887	0.270	0.903	0.146	**0.974**	0.002	0.380	**0.409**	0.886	0.253	0.904	0.000	0.015	0.004	0.957
Vehicle	0.320	0.975	0.303	0.967	0.172	0.977	0.312	0.976	**0.335**	**0.990**	0.316	0.976	**0.335**	0.986	0.153	0.979	0.000	0.000
Expanding	**0.993**	**0.999**	0.953	0.969	0.844	0.927	0.895	0.900	0.938	0.970	0.350	0.543	0.756	0.868	0.000	0.005	0.035	0.932
Gradual	**0.998**	**1.000**	0.956	0.964	0.921	0.978	0.995	**1.000**	0.978	0.984	0.322	0.452	0.759	0.838	-0.000	0.003	0.690	0.972
Merge-Split	0.963	0.988	0.952	0.979	0.877	0.947	**0.985**	**0.999**	0.932	0.968	0.226	0.450	0.895	0.945	0.000	0.019	0.635	0.981
Sudden	**0.921**	**0.956**	0.904	0.939	0.878	0.947	0.855	0.884	0.774	0.841	0.255	0.442	0.646	0.757	-0.000	0.008	0.602	0.939
Fast-RBF	**0.999**	**1.000**	0.988	0.989	0.924	0.963	0.995	0.999	0.911	0.938	0.261	0.462	0.818	0.896	0.000	0.022	0.461	0.655
Mean Score	**0.622**	**0.955**	0.572	0.910	0.534	0.927	0.558	0.910	0.534	0.858	0.223	0.627	0.492	0.877	0.059	0.305	0.246	0.643
Mean Rank	**1.50**	**2.75**	3.90	5.25	4.70	4.20	3.90	3.60	4.55	4.40	6.50	7.25	4.75	5.40	8.05	6.90	7.15	5.25

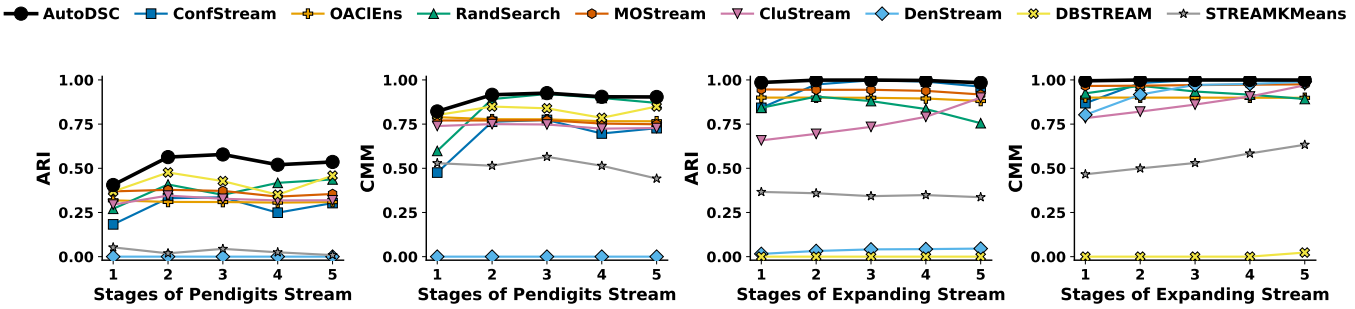
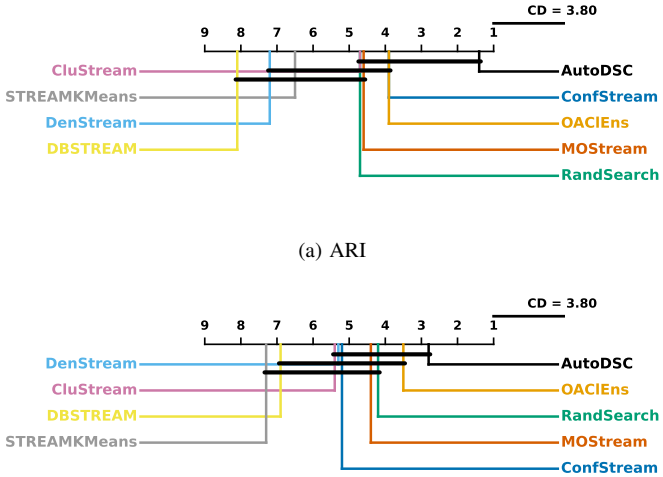


Fig. 3. Prequential clustering quality (ARI and CMM) across stream stages for AutoDSC and all baselines, on a representative real-world stream (Pendigits) and a synthetic drifting stream (Expanding). AutoDSC (thick black) stays at or above every baseline throughout the stream on both metrics.



(a) ARI

(b) CMM

Fig. 4. Critical Difference diagrams (Nemenyi, $\alpha = 0.05$) for ARI and CMM. Methods connected by a bar are not significantly different; AutoDSC holds the best mean rank on both metrics.

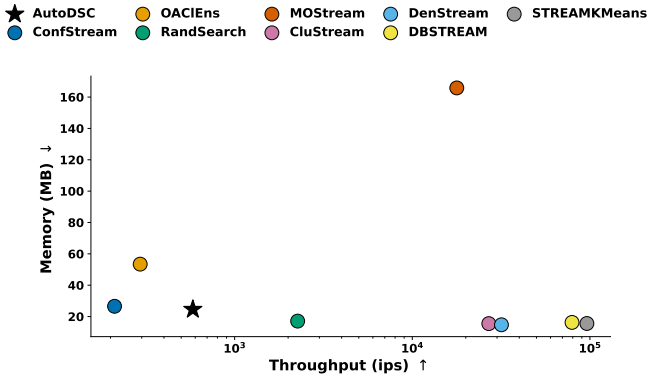


Fig. 5. Throughput vs. memory trade-off (mean over all datasets). AutoDSC (⋆) attains the highest clustering quality at a low memory footprint and moderate throughput.

computationally intensive offline Bayesian optimization [35]. While our current evaluation focuses on tabular data streams with numerical and categorical features, extensions to high-dimensional sparse text streams, multi-modal event streams, or image-based data would require domain-specific clustering algorithms beyond this scope. These novel contributions collectively bridge the gap between traditional AutoML approaches and the dynamic requirements of deploying real-time streaming applications in IoT, sensor networks, and analytics platforms, where manual configuration is impractical [17].

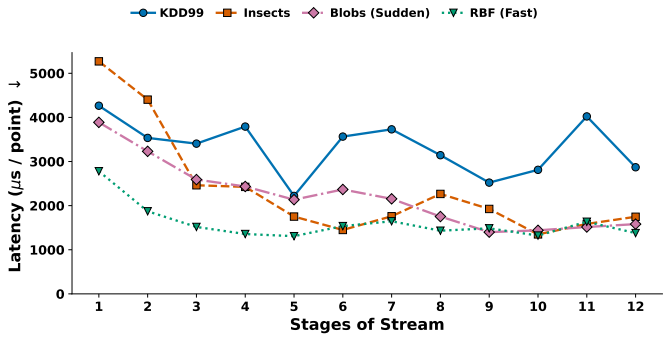


Fig. 6. Per-point latency across stream stages on representative real and synthetic streams. Latency is bounded and does not grow with stream length.

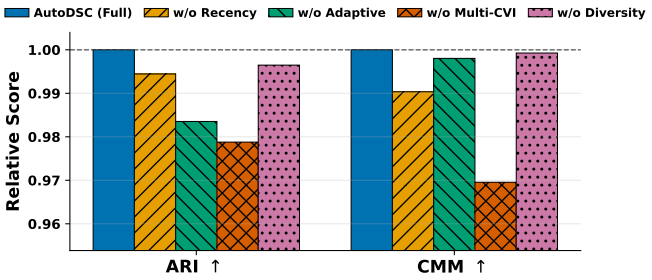


Fig. 7. Leave-one-out ablation on real-world streams: ARI and CMM of each variant relative to the full AutoDSC (dashed line = 1.0). Bars below the line indicate the removed component helps; the multi-objective CVI score contributes most.

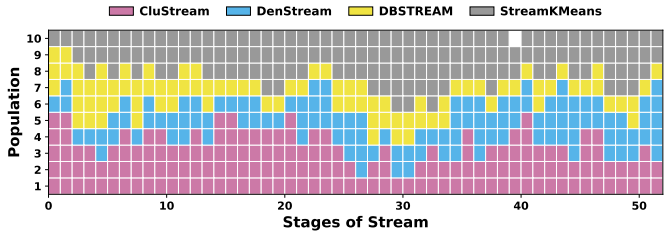


Fig. 8. Evolution of the AutoDSC population composition over the Insects stream. Rows are population slots, columns are stream stages, and colors denote the selected base algorithm.

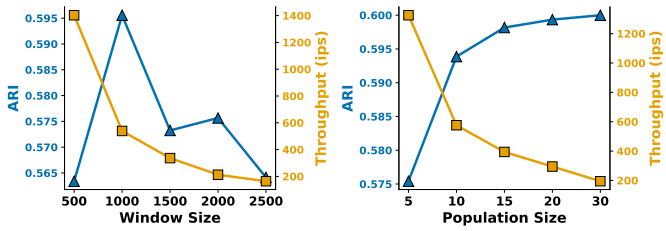


Fig. 9. Sensitivity of AutoDSC to window size (left) and population size (right): ARI (blue, left axis) and throughput (orange, right axis), averaged over all datasets. Defaults $w = 1000$, $K = 10$ sit at the quality/efficiency sweet spot.

VIII. Conclusion and Future Work

We presented AutoDSC, a novel AutoML framework for adaptive data stream clustering that unifies algorithm selection, hyperparameter optimization, and concept drift adaptation. Extensive experiments across diverse datasets demonstrate AutoDSC's statistical significance, achieving the highest average Rand Index while executing faster than existing frameworks. The framework is supported by theoretical analysis providing convergence guarantees in non-stationary environments. Future research directions include extending AutoDSC to semi-supervised settings, multi-task clustering, and advanced multi-objective optimization techniques to balance trade-offs such as accuracy, efficiency, and robustness in dynamic settings.

References

[1] F. Hutter, L. Kotthoff, and J. Vanschoren, *Automated machine learning: methods, systems, challenges*. Springer Nature, 2019.

[2] Y. Poulakis, C. Doulkeridis, and D. Kyriazis, "A survey on automl methods and systems for clustering," *ACM Transactions on Knowledge Discovery from Data*, vol. 18, no. 5, pp. 1–30, 2024.

[3] D. Tschechlov, M. Fritz, and H. Schwarz, "Automl4clust: Efficient automl for clustering analyses," in *Proceedings of the 24th International Conference on Extending Database Technology (EDBT)*. OpenProceedings.org, 2021, pp. 343–348.

[4] Y. Liu, S. Li, and W. Tian, "Autocluster: Meta-learning based ensemble method for automated unsupervised clustering," in *Advances in Knowledge Discovery and Data Mining (PAKDD 2021)*. Springer, 2021, pp. 246–258.

[5] R. ElShawi, H. Lekunze, and S. Sakr, "csmartml: A meta learning-based framework for automated selection and hyperparameter tuning for clustering," in *2021 IEEE International Conference on Big Data (Big Data)*. IEEE, 2021, pp. 1119–1126.

[6] D. Treder-Tschechlov, M. Fritz, H. Schwarz, and B. Mitschang, "Ml2dac: Meta-learning to democratize automl for clustering analysis," *Proc. ACM Manag. Data*, vol. 1, no. 2, Jun. 2023. [Online]. Available: https://doi.org/10.1145/3589289

[7] R. ElShawi and S. Sakr, "Tpe-autoclust: A tree-based pipline ensemble framework for automated clustering," in *2022 IEEE International Conference on Data Mining Workshops (ICDMW)*, 2022, pp. 1144–1153.

[8] M. C. da Silva, G. M. Tavares, E. Medvet, and S. B. Junior, "Problem-oriented automl in clustering," *arXiv preprint arXiv:2409.16218*, 2024, arXiv:2409.16218.

[9] L. Yang and A. Shami, "Iot data analytics in dynamic environments: From an automated machine learning perspective," *Engineering Applications of Artificial Intelligence*, vol. 116, p. 105366, 2022. [Online]. Available: https://www.sciencedirect.com/science/article/pii/S0952197622003803

[10] A. Bifet, R. Gavaldà, G. Holmes, and B. Pfahringer, *Machine Learning for Data Streams with Practical Examples in MOA*. MIT Press, 2018. [Online]. Available: https://moa.cms.waikato.ac.nz/book/

[11] B. Celik, P. Singh, and J. Vanschoren, "Online automl: An adaptive automl framework for online learning," *Machine Learning*, vol. 112, no. 6, pp. 1897–1921, 2023.

[12] N. Verma, A. Bifet, B. Pfahringer, and M. Bahri, "Asml: A scalable and efficient automl solution for data streams," in *Proceedings of the Third International Conference on Automated Machine Learning*, ser. Proceedings of Machine Learning Research, K. Eggensperger, R. Garnett, J. Vanschoren, M. Lindauer, and J. R. Gardner, Eds., vol. 256. PMLR, 09–12 Sep 2024, pp. 11/1–26. [Online]. Available: https://proceedings.mlr.press/v256/verma24a.html

[13] Q. Wu, C. Wang, J. Langford, P. Mineiro, and M. Rossi, "Chacha for online automl," in *Proceedings of the 38th International Conference on Machine Learning*, ser. Proceedings of Machine Learning Research, M. Meila and T. Zhang, Eds., vol. 139. PMLR, 18–24 Jul 2021, pp. 11 263–11 273. [Online]. Available: https://proceedings.mlr.press/v139/wu21d.html

[14] N. Verma, A. Bifet, B. Pfahringer, and M. Bahri, "Auto-reg: A dynamic automl framework for streaming regression," in *Pacific-Asia Conference on Knowledge Discovery and Data Mining*. Springer, 2025, pp. 245–256.

[15] M. Carnein, H. Trautmann, A. Bifet, and B. Pfahringer, "confstream: Automated algorithm selection and configuration of stream clustering algorithms," in *International conference on learning and intelligent optimization*. Springer, 2020, pp. 80–95.

[16] R. El Shawi and D. Rozgonjuk, "Onlineautoclust: A framework for online automated clustering," in *Proceedings of the 32nd ACM International Conference on Information and Knowledge Management*, 2023, pp. 3870–3874.

[17] M. Bahri, A. Bifet, J. Gama, H. M. Gomes, and S. Maniu, "Data stream analysis: Foundations, major tasks and tools," *Wiley Interdisciplinary Reviews: Data Mining and Knowledge Discovery*, vol. 11, no. 3, p. e1405, 2021.

[18] C. C. Aggarwal, "Data streams: models and algorithms," *Springer Science & Business Media*, 2013.

[19] C. C. Aggarwal, S. Y. Philip, J. Han, and J. Wang, "A framework for clustering evolving data streams," in *Proceedings 2003 VLDB conference*. Elsevier, 2003, pp. 81–92.

[20] F. Cao, M. Ester, W. Qian, and A. Zhou, "Density-based clustering over an evolving data stream with noise," in *SIAM International Conference on Data Mining (SDM)*, 2006. [Online]. Available: https://epubs.siam.org/doi/abs/10.1137/1.9781611972740.9

[21] M. Hahsler and M. Bolaños, "Clustering data streams based on shared density between micro-clusters," *IEEE Transactions on Knowledge and Data Engineering*, 2016. [Online]. Available: https://ieeexplore.ieee.org/document/7425260

[22] D. Sculley, "Web-scale k-means clustering," in *WWW 2010, Raleigh, North Carolina, USA*, 2010. [Online]. Available: http://code.google.com/p/sofia-ml

[23] J. Montiel, M. Halford, S. M. Mastelini, G. Bolmier, R. Sourty, R. Vaysse, A. Zouitine, H. M. Gomes, J. Read, T. Abdessalem *et al.*, "River: machine learning for streaming data in python," *Journal of Machine Learning Research*, vol. 22, no. 110, pp. 1–8, 2021.

[24] L. O'Callaghan, N. Mishra, A. Meyerson, S. Guha, and R. Motwani, "Streaming-data algorithms for high-quality clustering," in *Proceedings of the 18th International Conference on Data Engineering (ICDE'02)*, 2002. [Online]. Available: http://www-db.stanford.edu/~loc/icde02.ps

[25] P. Auer, N. Cesa-Bianchi, and P. Fischer, "Finite-time analysis of the multiarmed bandit problem," *Machine learning*, vol. 47, no. 2-3, pp. 235–256, 2002.

[26] N. Cesa-Bianchi and G. Lugosi, *Prediction, learning, and games*. Cambridge university press, 2006.

[27] O. Besbes, Y. Gur, and A. Zeevi, "Stochastic multi-armed-bandit problem with non-stationary rewards," in *Advances in neural information processing systems*, 2014, pp. 199–207.

[28] W. M. Rand, "Objective criteria for the evaluation of clustering methods," *Journal of the American Statistical association*, vol. 66, no. 336, pp. 846–850, 1971.

[29] H. Kremer, P. Kranen, T. Jansen, T. Seidl, A. Bifet, G. Holmes, and B. Pfahringer, "An effective evaluation measure for clustering on evolving data streams," in *Proceedings of the 17th ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, ser. KDD '11. New York, NY, USA: Association for Computing Machinery, 2011, pp. 868–876. [Online]. Available: https://doi.org/10.1145/2020408.2020555

[30] J. Demšar, "Statistical comparisons of classifiers over multiple data sets," *Journal of Machine learning research*, vol. 7, no. Jan, pp. 1–30, 2006.

[31] J. Bergstra and Y. Bengio, "Random search for hyper-parameter optimization," *The journal of machine learning research*, vol. 13, no. 1, pp. 281–305, 2012.

[32] Z. Wang, X. Wang, and S. Zhang, "MOStream: A modular and self-optimizing data stream clustering algorithm," in *2024 IEEE International Conference on Data Mining (ICDM)*, 2024, pp. 500–509.

[33] O. Arbelaitz, I. Gurrutxaga, J. Muguerza, J. M. Pérez, and I. Perona, "An extensive comparative study of cluster validity indices," *Pattern recognition*, vol. 46, no. 1, pp. 243–256, 2013.

[34] Y. Liu, Z. Li, H. Xiong, X. Gao, and J. Wu, "Understanding of internal clustering validation measures," in *2010 IEEE International Conference on Data Mining*, 2010, pp. 911–916.

[35] F. Hinder, A. Artelt, and B. Hammer, "Towards non-parametric drift detection via dynamic adapting window independence drift detection (dawidd)," in *International Conference on Machine Learning*. PMLR, 2020, pp. 4249–4259.